



Project Title: Knowledge Graph for Analyzing Housing Prices and Accessibility in Vienna

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Mode: ☒ 6 ECTS / ☐ 3 ECTS

Representations

(LO1) Understand and apply Knowledge Graph Embeddings	☐ I showed basic proficiency ☒ exceeded basic proficiency
I created a simple embedding-style representation by converting districts into normalized feature vectors and applying cosine similarity analysis to identify similar districts.	Pages 10–11
(LO2) Understand and apply logical knowledge in KGs	☐ I showed basic proficiency ☐ exceeded basic proficiency
Not addressed in this portfolio.	N/A
(LO3) Understand and apply Graph Neural Networks	☐ I showed basic proficiency ☐ exceeded basic proficiency
Graph Neural Networks were not used in this project.	N/A
(LO4) Compare different Knowledge Graph data models from the database, semantic web, machine learning and data science communities.	☐ I showed basic proficiency ☒ exceeded basic proficiency
I designed a Knowledge Graph data model consisting of District, Station, and AccessibilityCategory entities and defined relationships between them.	Pages 4–5



Systems

(LO5) Design and implement architectures of a Knowledge Graph	☐ I showed basic proficiency ☒ exceeded basic proficiency
I designed and implemented a complete Knowledge Graph architecture including schema design, graph construction, graph evolution, querying, and visualization.	Pages 4–10
(LO6) Describe and apply scalable reasoning methods in Knowledge Graphs	☒ I showed basic proficiency ☐ exceeded basic proficiency
I derived accessibility scores and accessibility categories from graph relationships, demonstrating graph-based inference and derived knowledge generation.	Pages 9–10
(LO7) Apply a system to create a Knowledge Graph	☐ I showed basic proficiency ☒ exceeded basic proficiency
I created a complete Knowledge Graph using Python, Pandas, and NetworkX from multiple datasets and performed graph-based analytical queries.	Pages 5–7
(LO8) Apply a system to evolve a Knowledge Graph	☐ I showed basic proficiency ☒ exceeded basic proficiency
I evolved the graph by introducing derived accessibility categories, additional nodes, and new semantic relationships.	Pages 9–10



Applications

(LO9) Describe and design real-world applications of Knowledge Graphs	☐ I showed basic proficiency ☒ exceeded basic proficiency
The project applies Knowledge Graph techniques to a real-world urban analytics scenario involving housing prices and public transport accessibility in Vienna.	Pages 3–12
(LO10) Describe financial Knowledge Graph applications	☐ I showed basic proficiency ☐ exceeded basic proficiency
Financial Knowledge Graph applications were not addressed in this portfolio.	N/A
(LO11) Apply a system to provide services through a Knowledge Graph	☐ I showed basic proficiency ☒ exceeded basic proficiency
The Knowledge Graph provides analytical services through graph queries, accessibility analysis, visualizations, and similarity analysis.	Pages 5–11
(LO12) Describe the connections between Knowledge Graphs (KGs), Machine Learning (ML) and Artificial Intelligence (AI)	☐ I showed basic proficiency ☒ exceeded basic proficiency
The project demonstrates how Knowledge Graphs can be connected with data science workflows and machine learning concepts through feature engineering and vector-based similarity analysis.	Pages 10–12



Additional Information

HAS NO EFFECT ON MARKING!

(please fill it out honestly, even if it is less than what is suggested in the ECTS breakdown – you are not judged on time spent!)

How many hours did you spend on your mini-project ?	65 hours
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* please exclude any hours you spent on parts reused from other courses

How many hours did you spend on your portfolio document preparation (this PDF) ?	15 hours
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* please exclude any hours you spent on parts reused from other courses

Declaration

I have marked all parts generated by Generative AI (e.g., ChatGPT) or otherwise copied from other sources and given any prompt I used either in a footnote or in an appendix making clear which parts are generated by which prompts or similar.	☒ I confirm this
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Please indicate if you have used generative AI parts of the mini-project	☒ I used it for roughly 20 % of the mini-project
Generative AI was used primarily for brainstorming, debugging support, code review, and documentation assistance. The Knowledge Graph design, dataset preparation, graph construction, analysis, visualizations, graph evolution, and final implementation were completed, verified, and adapted by the author.	

Please indicate if you have used generative AI parts of this document	☒ I used it for roughly 40 % of this document
Generative AI was used to improve wording, grammar, structure, and report drafting. All technical content, results, figures, interpretations, and learning outcome mappings were reviewed, validated, and finalized by the author.	

